

\* Cluster

$$A1 = (2, 10), A2 = (2, 5), A3 = (8, 4)$$

$$A4 = (5, 8), A5 = (7, 5), A6 = (6, 4), A7 = (1, 2), A8 = (4, 1)$$

$$C1 = A1 = (2, 10), C2 = A3 = (8, 4)$$

$$C3 = A7 = (1, 2)$$

→ Assign Points to the nearest centroid:

$$\sqrt{(x1 - x2)^2 + (y1 - y2)^2}$$

Distance to  $C1 (2, 10)$ :

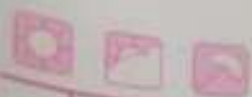
$$A1 = 0$$

$$A2(2, 5) = \sqrt{0 + 25} = 5 \quad A6(6, 4) = \sqrt{16 + 36} = 7.21$$

$$A3(8, 4) = \sqrt{36 + 36} = 8.485 \quad A7(1, 2) = \sqrt{1 + 64} = 8.06$$

$$A4(5, 8) = \sqrt{9 + 4} = 3.6 \quad A8(4, 1) = \sqrt{4 + 1} = 2.23$$

$$A5(7, 5) = \sqrt{25 + 25} = 7.07$$



- Distance to  $C_2(8,4)$ :

$$A_1(2,10) = \sqrt{36+36} = 8.485$$

$$A_2(2,5) = \sqrt{36+1} = 6.082$$

$$A_3(8,4) = 0$$

$$A_6(6,4) = \sqrt{4+0} = 2$$

$$A_4(5,8) = \sqrt{9+16} = 5$$

$$A_7(1,2) = \sqrt{49+4} = 7.28$$

$$A_5(7,5) = \sqrt{2+1} = 1.414$$

$$A_8(4,9) = \sqrt{16+25} = 6.40$$

- Distance to  $C_3(1,2)$ :

$$A_1(2,10) = \sqrt{1+64} = 8.06$$

$$A_2(2,5) = \sqrt{1+9} = 3.16$$

$$A_3(8,4) = \sqrt{49+4} = 7.28$$

$$A_6(6,4) = \sqrt{25+4} = 5.385$$

$$A_4(5,8) = \sqrt{16+36} = 7.21$$

$$A_7(1,2) = \sqrt{0+0} = 0$$

$$A_5(7,5) = \sqrt{36+9} = 6.71$$

$$A_8(4,9) = \sqrt{9+49} = 7.62$$



\* Compare the distance between each point and the 3 selected group:

Point A1 to the cluster → C1

Point A2 to the cluster → C3

Point A3 to the cluster → C2

Point A4 to the cluster → C1

Point A5 to the cluster → C2

Point A6 to the cluster → C2

Point A7 to the cluster → C3

Point A8 to the cluster → C1

- Calculate mean of cluster 1:

$$X = (2 + 5 + 4) / 3 = 3.67$$

$$\text{New } C1 = (3.67, 9)$$

$$Y = (10 + 8 + 9) / 3 = 9$$

- Mean of cluster 2:

$$X = (8 + 7 + 6) / 3 = 7$$

$$\text{New } C2 = (7, 4.33)$$

$$Y = (4 + 5 + 4) / 3 = 4.33$$

- Mean of cluster 3:

$$X = (2+1)/2 = 1.5$$

New  $C_3 = (1.5, 3.5)$

$$Y = (5+2)/2 = 3.5$$

\* Recalculate the distance from each point to the cluster means:

- Distance to  $C_1 (3.67, 9)$ :

$$A_1 (2, 10) = \sqrt{2.79 + 1} = 1.86$$

$$A_2 (2, 5) = \sqrt{2.78 + 16} = 4.47$$

$$A_3 (8, 4) = \sqrt{18.75 + 25} = 6.61$$

$$A_4 (5, 8) = \sqrt{1.76 + 1} = 1.66$$

$$A_5 (7, 5) = \sqrt{21.1 + 16} = 5.20$$

$$A_6 (6, 4) = \sqrt{5.43 + 25} = 5.52$$

$$A_8 (4, 9) = \sqrt{0.12 + 0} = 0.33$$

$$A_7 (1, 2) = \sqrt{7.23 + 49} = 7.5$$



- Distance to C2 (7, 4.33):

$$A1(2, 10) = \sqrt{25 + 32.25} = 7.56$$

$$A2(2, 5) = \sqrt{25 + 0.45} = 5.05$$

$$A3(8, 4) = \sqrt{1 + 0.12} = 1.05 \quad A6(6, 4) = \sqrt{1 + 0.12} = 1.05$$

$$A4(5, 8) = \sqrt{4 + 13.5} = 4.18 \quad A7(1, 2) = \sqrt{36 + 5.43} = 6.44$$

$$A5(7, 5) = \sqrt{0 + 0.45} = 0.67 \quad A8(4, 9) = \sqrt{9 + 21.81} = 5.55$$

- Distance to C3 (1.5, 3.5):

$$A1(2, 10) = \sqrt{(2-1.5)^2 + (10-3.5)^2} = 6.52$$

$$A2(2, 5) = \sqrt{(2-1.5)^2 + (5-3.5)^2} = 1.58$$

$$A3(8, 4) = 6.53 \quad A6(6, 4) = \sqrt{(6-1.5)^2 + (4-3.5)^2} = 4.55$$

$$A4(5, 8) = 5.70 \quad A7(1, 2) = \sqrt{(1-1.5)^2 + (2-3.5)^2} = 1.58$$

$$A5(7, 5) = 5.70 \quad A8(4, 9) = \sqrt{(4-1.5)^2 + (9-3.5)^2} = 6.10$$



\* Compare the distance between each point and the 3 selected group:

$A1, A4, A8 \rightarrow C1$

$A2, A7 \rightarrow C3$

$A3, A5, A6 \rightarrow C2$

- Mean of Cluster 1:  $(3.67, 9)$

- Mean of Cluster 2:  $(7, 4.33)$

- Mean of Cluster 3:  $(1.5, 3.5)$

→ The mean of clusters does not change compared to step 2, so it's the final solution.

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## ② Initial centroids:

$$C1 = (1, 1) \quad C2 = (5, 7) \quad C3 = (3, 4)$$

-Distance to  $C1 (1, 1)$ :

$$A1(1, 1) = 0$$

$$A2(1.5, 2) = \sqrt{(1.5-1)^2 + (2-1)^2} = 1.12$$

$$A3(3, 4) = \sqrt{(3-1)^2 + (4-1)^2} = 3.61$$

$$A4(5, 7) = \sqrt{(5-1)^2 + (7-1)^2} = 7.21$$

$$A5(3.5, 5) = \sqrt{(3.5-1)^2 + (5-1)^2} = 5$$

$$A6(4.5, 5) = \sqrt{(4.5-1)^2 + (5-1)^2} = 5.65$$

$$A7(3.5, 4.5) = \sqrt{(3.5-1)^2 + (4.5-1)^2} = 4.72$$

-Distance to  $C2 (5, 7)$ :

$$A2(1, 1) = \sqrt{(1-5)^2 + (1-7)^2} = 7.21$$

$$A2(1.5, 2) = 6.12$$

$$A3(3, 4) = 4.47$$

$$A4(5, 7) = 0$$

$$A5(3.5, 5) = 2.5$$

$$A6(4.5, 5) = 2.06$$

$$A7(3.5, 4.5) = 2.5$$

Distance to  $C3(3, 4)$ :

$$A1(1, 1) = 3.61$$

$$A2(1.5, 2) = 2.5$$

$$A3(3, 4) = 0$$

$$A4(5, 7) = 4.24$$

$$A5(3.5, 5) = 1.12$$

$$A6(4.5, 5) = 2.80$$

$$A7(3.5, 4.5) = 0.71$$

\* Cluster Assigne:

$$\underline{A1, A2} \Rightarrow C1$$

$$\underline{A4, A6} \Rightarrow C2$$

$$\underline{A3, A5, A7} \Rightarrow C3$$

$$\text{mean of } C1 = ((1+1.5)/2, (1+2)/2) = (1.25, 1.5)$$

$$\text{mean of } C2 = ((5+4.5)/2, (7+5)/2) = (4.75, 6)$$

- mean of  $C_3 = ((3+3.5+3.5)/3, (4+5+4.5)/3)$   
 $= (3.33, 4.5)$

\*Now we should recalculate the distance from each point to the new cluster mean, but I ask ChatGPT to do that and give me final result.

→ cluster 1 ( $C_1$ ) :→  $A_1, A_2$

→ cluster 2 ( $C_2$ ) :→  $A_4, A_6$

→ cluster 3 ( $C_3$ ) :→  $A_3, A_5, A_7$

⇒ Same assigne to each cluster compared to step 2, so it's the Final solution. #

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## LAB 2

Example: The Patient has the Flu?

$$P(\text{Flu} = \text{yes}) = 5/8 = \boxed{0.625}$$

$$P(\text{Flu} = \text{no}) = 3/8 = \boxed{0.375}$$

$$P(\text{Chills} = Y \mid \text{hasFlu} = \text{yes}) = 3/5 = \boxed{0.6}$$

$$P(\text{Chills} = Y \mid \text{hasFlu} = \text{no}) = 1/3 = \underline{0.3}$$

$$P(\text{runny nose} = N \mid \text{hasFlu} = \text{yes}) = 1/5 = \boxed{0.2}$$

$$P(\text{runny nose} = N \mid \text{hasFlu} = \text{no}) = 2/3 = \underline{0.67}$$

$$P(\text{headache} = \text{Mild} \mid \text{hasFlu} = \text{yes}) = 2/5 = \boxed{0.4}$$

$$P(\text{headache} = \text{Mild} \mid \text{hasFlu} = \text{no}) = 1/3 = \underline{0.3}$$

$$P(\text{fever} = Y \mid \text{hasFlu} = \text{yes}) = 4/5 = \boxed{0.8}$$

$$P(\text{fever} = Y \mid \text{hasFlu} = \text{no}) = 1/3 = \underline{0.3}$$

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$$P(X | \text{hasFlu} = \text{yes}) =$$

$$0.6 \times 0.2 \times 0.4 \times 0.8$$

$$= \boxed{0.0384}$$

$$P(X | \text{hasFlu} = \text{no}) =$$

$$0.3 \times 0.67 \times 0.3 \times 0.3$$

$$= \boxed{0.0181}$$

$$P(X | \text{hasFlu} = \text{yes}) P(\text{Flu} = \text{yes})$$

$$= 0.0384 \times 0.625 = \boxed{0.024} \Leftarrow$$

$$P(X | \text{hasFlu} = \text{no}) P(\text{Flu} = \text{no})$$

$$= 0.0181 \times 0.375 = \boxed{0.006787}$$

→ The naïve Bayes classifier predicts the  
Patient has the Flu. #

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